

EDOPS Data Characterization Report

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1. Purpose

EDOPS is the technical instrument of **Computing Place**, a research program of independent scholar Karl Grossner that aims to support systematic investigation of relationships between environmental setting and cultural phenomena. Computing Place is organized as complementary modules: an *Environmental Dimensions of Place Service (EDOPS)* and *Cultural Dimensions of Place (CDOP)*, in development). EDOPS produces *signatures*—structured, multi-band environmental descriptors of places, serialized as JSON¹, suitable for multiple research and educational purposes, including comparative analyses, classification experiments, and qualitative interpretation of CDOP cultural data.

An EDOPS signature is a compact document organized into persistence-banded layers (terrain and geology; hydrology and soil; climate; contemporary human geography; 'coastality'; and a temporal layer that includes paleoclimate reanalysis, land-use reconstruction, and volcanic forcing records). The signature draws on multiple source datasets—BasinATLAS² as the primary partition and attribute base, with LMR v2.1³, HYDE 3.4⁴, eVol2k v4⁵, and One Earth ecoregions⁶ integrated for temporal and ecological dimensions. The basin data carries both local (s) and upstream (u) values for many variables. The full design rationale is in the EDOP project summary documents (e.g. EDOP_summary_20260608).

The EDOPS design commitments depend on a foundational account of what each component variable *does*—its distributional properties, its spatial behavior, its scale sensitivity, its co-variance/concordance structure with respect to other variables, and its relative position within the range of its global values. The **characterization phase of EDOPS (termed CHAR)** has established that foundation with distinct **Exploratory Data Analysis (EDA)** and **Exploratory Spatial Data Analysis (ESDA)** tracks.

The CHAR phase had three operational outputs:

- An **variable codebook** (edops_codebook_v03.tsv), now augmented with per-variable position-attribute decisions, scale-sensitivity flags, historical-validity assignments, co-variant partners, and typology assignments. The codebook is the operational reference for EDOPS signatures
- A **set of methodological findings** about EDOPS variables' spatial and distributional attributes,

¹**JSON (JavaScript Object Notation)** is a widely adopted standard, human-readable text format for representing structured data as collections of key–value pairs and lists.

²BasinATLAS is the watershed component of HydroATLAS, providing standardized environmental, hydrological, and physiographic data for nested river basins worldwide. <https://www.hydrosheds.org/hydroatlas>

³<https://www.ncei.noaa.gov/access/paleo-search/study/27850>

⁴<https://hyde-portal.geo.uu.nl/>

⁵<https://b2find.eudat.eu/dataset/1b232535-4edd-5ef4-8acf-f23b733be685>

⁶<https://www.oneearth.org/bioregions-2023/>

summarized in this document and used to inform subsequent work (e.g. areal aggregation, correspondence experiments, worked examples, and a user-facing rubric for query design) cf. `eda_log-findings.md`, `esda_log-findings.md`.

- An **enumeration of open design decisions** flagged for resolution at the October expert review and in post-CHAR development work (§10 below).
- A **mapping addition to the EDOPS sandbox** (edops.kgeographer.org/sandbox) that displays a) individual variable distributions across the basin set at global and regional scales, and b) regional scatterplot comparisons of variable pairs. The interface serves as a visual companion to this document, the codebook and findings.

1.1 How to read this report

This draft report documents the Characterization (CHAR) phase of EDOPS, which established how each environmental variable behaves. Early sections describe what was characterized and how. In §2 the persistence bands that organize a signature are introduced. In §3 CHAR phase products are described, and §4 details one—the variable typology that tags each variable by the kind of spatial claim it supports.

The sections that follow explain what these characteristics mean for choosing signature parameters and for interpreting results. §5 *Reading CHAR Phase Results*, sets out three interpretive propositions that govern how a value should be understood: that spatial structure is not environmental character, that spatial concordance is not attribute-space concordance, and that the local/upstream (s/u) duality is a tail phenomenon. §6 *Scale: L6 vs L8 in Practice* gives practical guidance on when, and why, the resolution of a query changes what a signature reports. The remaining sections are reference: a per-band walkthrough (§7), an explicit statement of what CHAR does not establish (§8), codebook augmentations generated (§9), and open design decisions still under review (§10).

A reader who wants the practical payoff first—how to read a signature, and when the choice of scale matters—can go straight to §5 and §6 and treat §2–§4 as the apparatus those sections rely on; a reader who prefers to build the method up in order can read straight through.

2. Persistence Bands

EDOPS organizes signature variables into **persistence bands**—layers grouped by the timescale over which their values can be treated as approximately stable. The band structure can be read as a *temporal-persistence taxonomy*. Terrain and geology change on geological timescales; hydrology and soils on millennial-to-centennial scales; climate on centennial scales (as a baseline, with explicit complications for paleoclimate); human geography on the timescale of recent industrial transformation.

The bands as currently defined:

- **Band A—Terrain and geology.** Elevation, slope, lithology, karst extent, glacier and permafrost coverage. Stable on human timescales; defensible for any query period.
- **Band B—Hydrology and soil.** Discharge regimes, soil texture, runoff, groundwater, wetland classification, river network properties. Mostly stable on centennial scales; specific variables

(degree of regulation, contemporary river area) reflect modern infrastructure and are modern-only.

- **Band C—Climate.** Mean and seasonal temperature, precipitation, aridity, annual evapotranspiration (potential, PET; actual, AET), snow cover, biome and ecoregion classification. Sourced from contemporary measurement and treated as a centennial baseline; explicit qualifying notes apply for pre-modern queries. Absolute paleoclimate context for CE-period queries comes from Band T (LMR), not Band C.
- **Band D—Human geography.** Contemporary anthropogenic variables, including population density, human footprint, cropland extent, Gross Domestic Product (GDP), and Human Development Index (HDI). Useful for internal correspondence studies and some post-industrial research; not a baseline for earlier periods.
- **Band E—'Coastality'.** Currently, flow distance to marine outlet, and outlet type; future additions to include topological depth from coast, and shelf topography. Stable on geomorphological timescales (active deltas are the known exceptions); defensible for any query period.
- **Band T—Temporal.** Paleoclimate reanalysis (LMR, ~700–1900 CE reliable window), land-use reconstruction (HYDE, 10,000 BCE–present), and volcanic forcing catalog (eVol2k, ~500 BCE–1900 CE). The only band whose values vary by query time and location rather than by location alone. Per-source reliability constraints are addressed in §8.

The bands enable some temporal scoping of signature queries. A query for a BCE site can defensibly draw on Bands A, B, E, and the available range of Band T, while qualifying Band C as a contemporary baseline used in lieu of unavailable paleoclimate data, and suppressing or down-weighting Band D. A modern query can engage all bands without qualification. Users can request band-subset signatures at query time, or retain the full signature and apply down-weighting at analysis time; in either case, qualifying notes in the codebook and a `_note` field in JSON payloads inform users of variables' epistemic status.

Two overall qualifications must be emphasized: first, many major river basins have been substantively modified across the last few millennia. Therefore, Band B values in such regions are not accurate pre-human baselines. The EDOPS persistence framing supports query scoping and does not claim pre-modern variables capture pre-modern conditions exactly. Furthermore, landscape change trajectories—the historical processes by which contemporary values arose—are outside the scope of EDOPS signatures; what Bands A-E report is the contemporary state, and together with Band T variables indicate how stably that state can be projected backward.

The inclusion of *Band D-Anthropocene markers* is justified by their value as future targets for planned cultural-environmental correspondence analyses with variables in other bands and external datasets. CHAR has produced internal correspondence findings within Band D indicating spatial coherence, e.g. $HDI \times GDP$ asymmetry, where high-HDI low-GDP enclaves number 8,566 basins against 7 in the reverse configuration, and those enclaves lying within plausible regions such as former Soviet

states. The banded structure facilitates suppression of Band D for queries where it is not interpretively appropriate.

3. CHAR Phase Products

CHAR encompassed two coordinated strands of work that together produced the augmented codebook and this document. The **Exploratory Data Analysis (EDA)** strand covered eleven numbered tasks producing distributional, correlational, missing-data, local/upstream divergence, geographic pre-clustering, and sampling-bias characterizations for all static bands (A–E) and the Band T variables. The **Exploratory Spatial Data Analysis (ESDA)** strand covered systematic spatial-statistics work at two BasinATLAS Pfafstetter levels (L6, ~16,000 basins; L8, ~190,000 basins): **univariate global Moran's I^7** and **local indicators of spatial association (LISA)** for 40+ Band A–D variables, bivariate Moran's I and LISA for selected variable pairs, categorical **join-count⁸** coherence for three categorical variables, and Band T spatial characterization at native grid resolution. Outputs from both strands are integrated into the augmented codebook (§9), the methodological findings developed in §§4–6, and the design-decisions enumeration in §10.

3.1 The LISA Classification

Where global Moran's I summarizes spatial autocorrelation across all terrestrial sub-basins in a single value, LISA decomposes it to the local level: each location receives a classification—HH (high value surrounded by high-value neighbors), LL (low surrounded by low), HL (high surrounded by low), LH (low surrounded by high)—with an associated significance test.

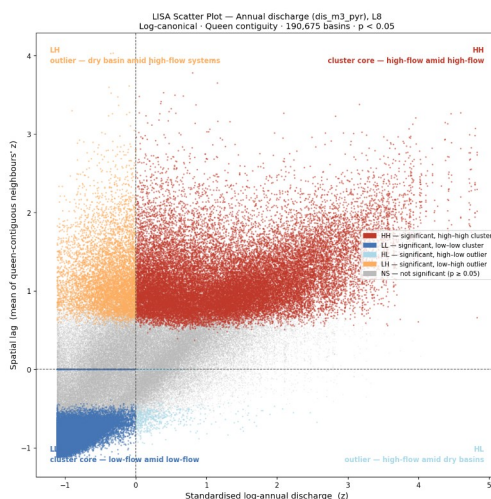


Figure 3.1. LISA scatterplot for log-transformed global annual discharge at L8: each basin plotted by its standardized value (x) against the mean of its queen-neighbors' standardized values (y). Points colored by significant LISA class ($p < 0.05$, 999 permutations); gray = NS. The slope through the colored points equals global Moran's I .

⁷ **Moran's I** is a measure of global spatial autocorrelation: how similar a variable's values are at neighboring locations relative to chance. Values near +1 indicate strong clustering of similar values; 0 indicates spatial randomness; negative values indicate dispersion (high-value locations tend to neighbor low-value locations).

⁸ **Join-count** statistics apply to categorical variables: they test whether category-boundaries occur more or less frequently than chance expectation. High "same-category neighbor" counts indicate spatial clustering of the category; high "different-category neighbor" counts indicate spatial dispersion or mixing.

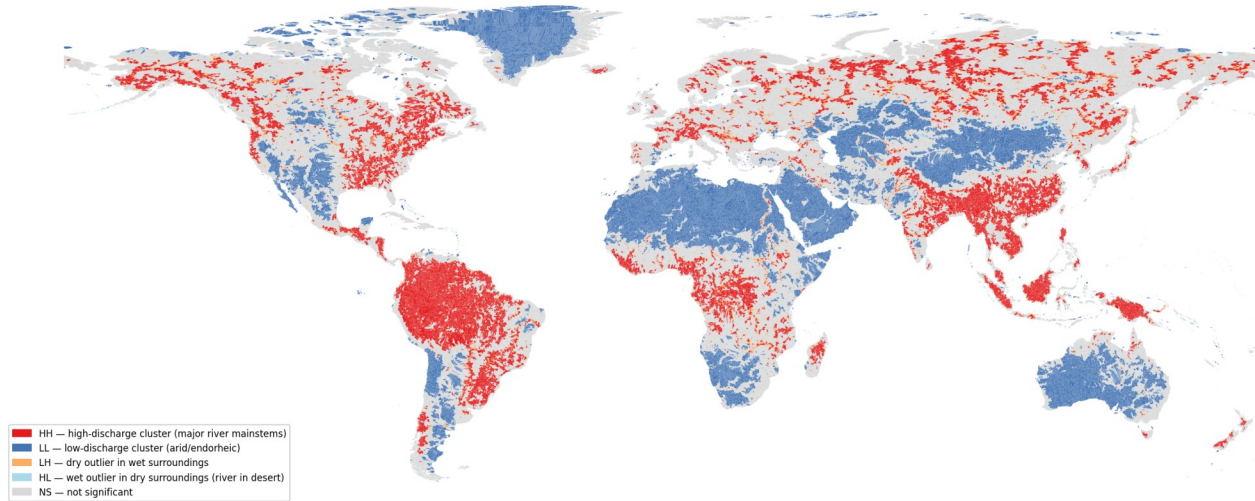


Figure 3.2. Global LISA classification of annual discharge at L8. HH cluster cores (red) follow major river mainstems; LL cluster cores (blue) coincide with arid and endorheic regions. Outliers (HL, LH) are present but not resolvable at global extent—see Figure 3.X (scatterplot) for the classification mechanism.

3.2 Output

The work has produced five categories of output. Each transfers differently to neighborhood-scale use of the instrument.

Variable-level properties. Each EDOPS variable has received a typology assignment (*continental-gradient*, *network-topology*, *scale-dependent*, or *local-anomaly*) and a scale-sensitivity flag indicating how its spatial coherence changes between L6 and L8. These are intrinsic properties of the variable rather than of the specific dataset, and they are directly usable as priors when interpreting a signature at any scale. They are the most transportable CHAR output. The typology is developed in §4; scale behavior is consolidated in §6.

Position-attribute calibration. The signature's globally-calibrated position descriptors—percentile, log-percentile, rarity rank, dominance class—require each variable's full global distribution to be characterized for the position to mean anything. CHAR provides that calibration; the per-variable position method and its rationale are recorded in the `position_method` and `position_notes` codebook columns. A user reading that a site's aridity sits at the 98th percentile globally receives a globally-anchored statement that depends on this calibration to be interpretable.

Regional ESDA case studies. A bounded set of regional bivariate studies—temperature × precipitation sign-reversal in the Mediterranean, elevation × slope behavior in the Tibetan and African Plateau cases, discharge-cluster regional behavior in cold-climate headwaters and monsoon-fed flash systems—characterize how specific variable pairs decouple from their global structure in specific regions. These studies are themselves neighborhood-scale findings and are directly transportable as interpretive guidance for queries in the regions concerned.

Sampling-bias and correspondence-power findings. Candidate cultural test datasets were characterized for their geographic sampling biases relative to the global basin population. The D-

PLACE anthropological dataset over-samples tropical wet mountains by a factor of 3.65; a World Heritage Cities corpus used in early CDOP investigations over-samples regulated river corridors by 5.55; both datasets share a cold/hyper-arid blind spot. These findings bound what subsequent correspondence experiments can claim and inform the design of the future areal aggregations phase.

Descriptive global statistics. Global Moran's I values, distributional summaries, missing-data patterns, and global co-variance structure are produced for completeness and as an empirical foundation for the position-attribute calibration. At regional scales, the directly applicable information is the typology assignment ("temperature is continental-gradient and spatially smooth at the scale of regional queries"), not the underlying global statistic. Global statistics support the typology assignment but are not themselves the interpretive output. The code and raw output of these analyses will be available in the EDOPS GitHub repository (<https://github.com/kgeographer/edops>).

In this document, variable typology is found in §4, three cross-cutting methodological propositions in §5, scale findings in §6, the per-band reference in §7, what CHAR does not establish in §8, the codebook in §9, and a design-decisions table in §10.

4. The Variable Typology

The variable typology is a key empirical output of the CHAR phase ESDA work. It assigns each of the 40 continuous-numeric variables from the BasinAtlas dataset to one of four classes based on global spatial-autocorrelation signature, telling users what kind of spatial claim the variable supports: locating a basin within a **continental-gradient**, within a hydrological **network-topology**, within a population of geographic outliers (**local-anomaly**), or having a **scale-dependent** spatial structure. The class assignment is recorded in the codebook's `typology_cluster` column. The typology—like the codebook columns and design positions reported elsewhere in this document—is subject to refinement through subsequent evaluation phases and expert scrutiny.

Of the 40 variables, 26 are currently implemented in the EDOPS API and 14 are planned. The planned variables were included in the CHAR sweep so the characterization findings could inform signature-inclusion decisions; this screening is flagged as an open task in §10. The broader EDOPS signature design also includes categorical variables (lithology, biome, ecoregion, wetland class—characterized via join-count statistics in §3) and Band T time-conditional variables (LMR paleoclimate, HYDE historical land use, eVolv2k volcanic forcing—characterized separately). The EDOPS codebook records implementation status for all variables.

4.1 Scope and diagnostics

The typology applies only to variables for which basin-level scalar Moran's I is mathematically appropriate and has been computed. Therefore, three classes of data are excluded:

- **Categorical variables** (lithology, biome, ecoregion, wetland class) are not on the same mathematical footing. Moran's I requires a scalar quantity to operate on; a categorical basin attribute (a biome label, say) cannot be autocorrelated in the gradient-shape sense. Categorical spatial behavior is characterized separately via join-count statistics, which test whether category boundaries respect chance expectation—a different methodology producing different diagnostics. Categoricals are addressed in §7 below.

- **Band T variables** from the LMR, HYDE, and eVolv2k datasets vary on time as well as space. The native spatial resolution for LMR and HYDE is gridded ($2^{\circ} \times 2^{\circ}$ and $5' \times 5'$ respectively) rather than basin-level. Computing Moran's I would require aggregating each variable for each time step to basin units and running the full sweep—a substantial effort flagged in the codebook as deferred-phase4. LMR and HYDE were characterized at native grid resolution and addressed in §7. The eVolv2k data is of events at point locations.
- **Band E variables** beyond `dist_sink_km` (outlet type, downstream topological depth) are derived measures that haven't been processed through the Moran's I machinery yet. They could be in a later sweep; outlet type may be better characterized as a categorical regardless.

Three diagnostics drive the class assignment for the 40 continuous-numeric variables:

- **Global Moran's I at L6 (I_{L6}):** the baseline measure of how strongly a variable's values cluster spatially at relatively coarse resolution. High I_{L6} (≥ 0.85) indicates that neighboring basins share values closely; low I_{L6} indicates either spatial randomness or spatial structure that L6 fails to resolve.
- **Scale direction ($\Delta_I = I_{L8} - I_{L6}$):** how spatial coherence changes when basin resolution is refined from L6 to L8. The dominant pattern (36 of 40 variables) is $\uparrow$: i.e. I increases at finer resolution because smaller basins sit more firmly inside their local climate or substrate zone. The minority pattern ($\downarrow$) is diagnostic of structurally different spatial organization.
- **Outlier prevalence:** the percentage of basins receiving HL or LH (spatial outlier) LISA classifications. The class diagnostic is whether the variable's *primary spatial signal* lives in scattered outliers rather than in broad gradients—distinct from the question of whether outlier basins of a given variable carry meaningful information, which can be true for any variable regardless of class. Most variables in the sweep show outlier fractions below 1%; one variable shows $>3\%$, crossing the operational threshold for the local-anomaly class.

The class boundaries are heuristic thresholds, not natural-kind boundaries. As such they are explicit candidates for expert review (§10).

4.2 The four classes

Class	Diagnostic	n	Mechanism	Exemplar variables
continental-gradient	$I_{L6} \geq 0.85$, scale direction $\uparrow$	21	Smooth large-scale gradients spanning hydrological boundaries	aridity, temperature, PET, AET, precipitation, soil texture, HDI, GDP, snow cover, runoff, cropland
network-topology	Scale direction $\downarrow$ ($\Delta_I < -0.010$)	3	Confluence value-jumps within drainage networks produce sharp adjacent-basin contrasts	annual discharge, minimum monthly discharge, degree of regulation

Class	Diagnostic	n	Mechanism	Exemplar variables
local-anomaly	Low I_L6, outlier prevalence > 3.0%	1	Spatial signal concentrated in scattered geographic outliers	local sub-basin river area (<i>ria_ha_ssu</i>)
scale-dependent	Moderate I_L6, ↑ direction, intermediate signature	15	Bounded features that smear at L6 and resolve as discrete features at L8	glaciers, karst, wetlands, erosion rate, slope, groundwater, human footprint

Full per-variable class assignments are in the `typology_cluster` column of the augmented codebook.

4.3 Class mechanisms

continental-gradient is the largest class, comprising most of Band C, Band A elevation, Band B soils and runoff, much of Band D, and Band E's *dist_sink_km*. High global Moran's I describes the *form* of spatial variation (smooth gradients spanning continents) rather than its magnitude. Continental-gradient variables span the largest dynamic ranges in the dataset—aridity from Atacama hyper-arid to Amazonian wet, temperature from polar to equatorial. The within-region variation visible at L8 is real and analytically important; the typology characterizes its global form, not its absence. The signature is high I_L6 (often ≥ 0.95) rising further at L8; PET, temperature, snow cover, aridity, and AET all reach $I_{L8} \geq 0.992$ —effectively maximum spatial autocorrelation. For these variables, the global pattern is fully resolved at L6 and L8 adds precision rather than new spatial structure. Two basins with similar continental-gradient values share a macro-environment; the variable is suitable for placing a basin in the global climate system.

network-topology is the smallest non-singular class, with three members: annual discharge, minimum monthly discharge, and degree of regulation. What places a variable here is that its value is accumulated or propagated along the drainage network rather than set by local geography — discharge sums all upstream catchment flow, minimum monthly discharge (baseflow) carries the same integration, and degree of regulation reflects upstream reservoir influence. The tag therefore tells a user that a basin's value expresses its position in the river network: two basins with similar network-topology values share a network position, not necessarily a climate or a location.

The diagnostic fingerprint is distinctive. Where continental-gradient variables have high Moran's I that rises at finer resolution, network-topology variables have lower I that holds flat or declines from L6 to L8 — minimum monthly discharge shows the sharpest decrease ($\Delta I = -0.041$). The mechanism is that discharge jumps one to two orders of magnitude at every confluence, where a small tributary meets a high-volume mainstem; at L6 these confluences are usually interior to a single sub-basin and invisible to Moran's I, while at L8 they fall on sub-basin boundaries, placing a low-discharge tributary queen-contiguous to a high-discharge mainstem — the LH outlier pattern (LH adjacencies grow 22.5× as basin count grows only 11.6×, DIS.6). For these variables, low or declining I is not noise but a positive signal of network organization, and a sharp neighbor contrast should be read as real confluence structure rather than error.

local-anomaly currently has a single member: local sub-basin river area (*ria_ha_ssu*)—a variable with near-zero spatial autocorrelation, whose neighboring basins don't predict one another. The

mechanism: most basins have small or near-zero local river area, but a scattered minority—typically small headwater basins where large channels surface, or major channel-corridor basins—carry disproportionately large values. The spatial signal lives in the outliers, not in gradients. A basin's position on this variable is most informative when the basin sits in the high-anomaly minority. The class is small but methodologically distinct: variables of this type require interpretation as flags of exceptional concentration, not as gradient positions. With currently only one member, the class is open to review—with a different outlier-prevalence threshold or a different diagnostic combination, *ria_ha_ssu* might fit better elsewhere (perhaps as a low-I scale-dependent variable), or the class slot might attract additional members as future variables are added or re-characterized. This is flagged for review at §10.

scale-dependent captures variables whose spatial form genuinely changes between L6 and L8—gradient-shaped at the coarse resolution, feature-shaped at the finer one. Glaciers, karst, wetlands, erosion rate (largest absolute scale gain in the entire sweep, $\Delta_I = +0.181$), groundwater depth, slope, and human footprint mostly fit this pattern. At L6 they look more like gradients than anomalies; at L8 they resolve as discrete features in defined geographic settings. The class label flags variables whose spatial behavior depends meaningfully on scale and whose interpretation should be informed by the scale of the query.

4.4 What the typology supports

Knowing a variable's class supports four concrete interpretive judgements:

Reading similarity between basins. Two basins with similar values for a *continental-gradient* variable share a macro-environment—similar climate or socioeconomic position. Two basins with similar values for a *network-topology* variable share a hydrological network position; they may sit in very different climate zones. For *scale-dependent* variables, similarity should be interpreted at the scale of the query: similar values at L8 indicate similar local feature presence; similar values at L6 indicate broader regional context. For *local-anomaly* variables, similarity in the high-value range identifies exceptional concentrations; similarity in the low-value range is geographically uninformative (most basins share the low-value background).

Predicting what a LISA map will look like. Class predicts spatial structure: continental-gradient produces smooth continent-scale HH and LL zones with limited spatial outliers; network-topology produces patterns aligned with drainage networks with notable LH spatial outliers; scale-dependent produces cluster-shaped maps at L6 and feature-shaped maps at L8; local-anomaly produces sparse spatial outliers in an otherwise uniform background.

Choosing scale for a query. Class informs the L6/L8 choice for a given use; this is detailed below.

Anticipating regional behavior. Continental-gradient variables may show regional decoupling—the Mediterranean temperature $\times$ precipitation sign-reversal, the Tibetan elevation $\times$ slope decoupling—where the global gradient pattern breaks down in specific regions. Network-topology variables tend to behave more consistently across regions because their structure is set by drainage geometry rather than climate. This conditions where regional ESDA case studies are most likely to be informative.

The typology does not answer two further questions:

- **Within-class co-variance.** Class membership describes how a variable is spatially structured, not how it relates to the other variables in its class. Within the *continental-gradient* class, two variables may co-vary closely across space or move almost independently, and the typology does not register the difference. That co-variation is a characterized property in its own right, treated in the bivariate concordance analysis (§5.2). Concordance is a relation between two variables, not shared information: each variable contributes a distinct measured dimension to the signature, and high concordance describes how two such dimensions move together — never grounds for treating either as redundant or dropping it. CHAR characterizes how variables co-vary; it does not screen them for removal.
- **Historical validity.** A variable's typology does not say whether the variable is appropriate for a given query period. Human Development Index HDI is *continental-gradient* and modern-only; PET is *continental-gradient* and a pattern-stable proxy for pre-modern queries. Typology and historical validity are independent axes that the augmented codebook records in separate columns.

4.5 User-facing implication: L6 vs L8

The typology directly informs the choice between L6 and L8 for a given query. Briefly:

- For **continental-gradient** variables, L6 and L8 are nearly interchangeable. The spatial pattern is fully resolved at L6; L8 adds basin-area precision but little new structural information.
- For **network-topology** variables, L8 sharpens the intra-drainage value contrasts described above—sub-basins along a mainstem register the cumulative flow of upstream catchments, while adjacent tributary sub-basins carry only their own. L6 averages across these contrasts. Use L8 to surface drainage discontinuities; use L6 for regional position summaries.
- For **scale-dependent** variables, L8 resolves many bounded features (glaciers, karst, wetlands) that are smeared across L6 basins. Use L8 for feature-presence queries; use L6 for regional context.
- For **local-anomaly** variables, L8 preserves the geographic precision of the anomaly; L6 averages it away.

§6 develops these implications systematically across the band structure and provides scale-selection guidance for representative use cases.

5. Reading CHAR Phase Results

Three propositions cut across the per-variable detail of §4 and condition how any EDOPS signature should be read. Each was established through the EDA and ESDA work of the characterization (CHAR) phase and reflect the structure of current EDOPS data components. They are presented here as methodological positions.

5.1. Spatial structure is not environmental character

A LISA class (HH, LL, HL, LH, NS) is assigned to each basin individually, but the class itself is defined by where the basin's value and its queen-neighbors' average value sit relative to

the *global* mean of the variable. LISA therefore identifies a basin's *position in the global distribution*, observed locally—not its absolute environmental character. Two basins can share a LISA class while occupying entirely different environments; two basins can occupy similar environments while sharing no LISA class.

Evidence. The Atacama Desert and the Argentine Pampas share an LL classification in the aridity $\times$ precipitation bivariate LISA (the aridity index increases with humidity and decreases with aridity—higher values are wetter). LL means the basin's value sits below the global mean and the average of its queen-neighbors' values also sits below the global mean; Atacama and Pampas both satisfy this on both metrics. But the Atacama is hyper-arid (P/PET well below 0.1) while the Pampas is a productive temperate grassland (P/PET $\approx$ 0.7), supporting one of the world's major agricultural regions. A signature using only LISA class would conflate them; they are spatially classed identically but environmentally distinct by an order of magnitude in aridity. The reason: "low" in LL refers only to the sign of two comparisons against one global threshold—magnitude, whether "extreme arid" or "moderately arid," does not enter. The complementary case also holds: two basins with similar environmental character but in different parts of the global distribution may receive different LISA classes.

Design implication. Signatures need both raw variable values *and* structural position. Raw values carry magnitude—the absolute environmental character of the basin. LISA class and percentile position carry structural information: where the basin sits relative to the global distribution, and how it sits relative to its queen-contiguous neighbors. Both are needed for meaningful comparison; neither suffices alone. The augmented codebook carries both: raw values appear in the signature payload alongside global-position attributes (percentile, log-percentile, rarity rank, dominance class) recorded per variable. The bivariate LISA classifications themselves are preserved at per-basin granularity in `output/edop/spatial/bivariate_redundancy.parquet` (2.1M rows) for downstream analyses that need them.

5.2. Spatial concordance is not attribute-space concordance

Two variables can be highly correlated globally and still carry distinct spatial information at basin level. The Pearson and Spearman correlations in EDA measure attribute-space concordance; bivariate Moran's I measures spatial concordance. High values in the first do not imply high values in the second.

Evidence. The discharge cluster is the clearest case. Annual discharge $\times$ maximum monthly discharge shows Pearson $r = 0.967$ —near-perfect attribute-space concordance. But bivariate Moran's I for the same pair is 0.538 at L8, which is *below* the univariate Moran's I of either variable (~ 0.563). The seasonal-regime geography that maximum discharge captures independently from annual mean is a spatial signal that the high global correlation does not detect. All three discharge pairs show this pattern—LH spatial-outlier counts substantially exceed HL counts ($dis_yr \times dis_min$: LH = 6,641 vs HL = 3,662; $dis_yr \times dis_max$: LH = 7,296 vs HL = 671). These LH basins are cold-climate or snowmelt-transition zones where small headwater catchments have low annual flow but neighbor great boreal perennial rivers, or monsoon-fed flash systems with regionally concentrated peak flows. The spatial structure is doing work that the high attribute-space correlation obscures.

The principle generalizes. Temperature $\times$ precipitation has a global bivariate Moran's I of +0.315—a moderate positive pattern reflecting the latitudinal climate gradient (the wet tropics are hot and rainy; polar regions are cold and dry). But within the Mediterranean basin, the same bivariate I is -0.250 : the global pattern sign-reverses regionally, because within Mediterranean basins warmer locations tend to be drier and cooler locations wetter. Temperature $\times$ snow cover shows very strong negative spatial coupling globally (bivariate Moran's I = -0.865)—cold neighborhoods have snow, warm ones do not—but collapses to not-significant in the Tibetan and cold-arid interior, where all basins are cold and snow distribution is governed by precipitation patterns independently.

Design implication. In predictive modeling, high pairwise correlation between candidate features often justifies dropping one to reduce dimensionality. EDOPS signatures are not a predictive model: no signature variable was removed on the basis of global co-variation. L8 basin signatures need local character even where two variables are highly correlated globally, because the local character is exactly what high global correlation obscures. The `high_r_partner` codebook column records which variables share global Spearman $|r| \geq 0.90$; it documents the correlation structure and does not authorize removal. For users who want to reduce a high-correlation pair for a specific analytical purpose, the column provides the information; the signature itself retains both variables.

5.3. The local/upstream (s/u) duality is a tail phenomenon

For each of several variables, the EDOPS signature reports both a basin-local value (s) and an upstream-cumulative value (u). This local/upstream duality is meaningful for only a minority of basins; in most cases, s and u are nearly identical and the dual reporting adds no information. However, where they diverge, the divergence is itself an analytically interesting finding. The s/u divergence is less *pervasive* than one might imagine but *more* diagnostic where it occurs.

Evidence. Across nine s/u variable pairs at L8, the global median divergence is exactly zero. The middle half of basins (the interquartile range, from the 25th to the 75th percentile) shows divergence at or near zero for six of the nine pairs. Strong divergence is concentrated in the tail of the distribution: notable values appear only beyond the 95th percentile, and the strongest cases beyond the 99th. The pattern indicates that most L8 sub-basins have upstream catchment area approximately equal to their local catchment area—they sit at headwater or near-headwater positions in the drainage hierarchy, with little or no upstream area cumulating from outside the sub-basin itself. The bivariate spatial confirmation is independent: HL+LH proportions for $s \times u$ bivariate LISA range from 0.2% (annual temperature, annual precipitation) to 0.6% (human footprint)—well under 1% of basins in each case. Both senses of "divergence"—per-basin distributional and spatial-bivariate—concur. The duality is informative for a small fraction of basins, not as a generic dataset property.

Two directional asymmetries are worth flagging within this proposition. Temperature is almost always upstream-colder where it diverges (91.8% of divergent basins; the 5th percentile of the divergence distribution is -3.13°C and the 1st percentile is -7.2°C). This is the cold-upstream signal of *exotic rivers* — rivers whose flow originates in distant wet uplands and crosses drier lowlands—like the Indus, the Tigris-Euphrates, and the Ganges. Human footprint is almost always greater locally than upstream (80–90% of basins), indicative of settlement and agriculture concentrating in lowland, accessible positions rather than headwaters. These asymmetries are themselves meaningful: where

divergence is present for temperature, it identifies a specific basin type (lowland fed by cold mountains); where it is present for human footprint, it identifies the settlement footprint itself relative to its hinterland.

Design implication. Where $S \approx u$, reporting both as separate values adds no information. Where they diverge, the divergence magnitude is the meaningful contribution. The signature should be read with this in mind: a small or zero s/u difference is the default for most basins; a non-trivial difference is the signal. Whether divergence magnitude should become a first-class signature feature alongside S and u raw values is flagged for §10.

A coordinated methodological position

The three propositions amount to a coordinated methodological position. The first holds that LISA class carries structural-position information but not environmental character—raw values must be read alongside it. The second holds that high global correlation between two variables does not imply spatial concordance; the decoupling cases carry information beyond what the global correlation captures. The third holds that while the s/u values nearly coincide for most basins; the meaningful divergence between local and upstream lives in the distributional tails.

Together, they suggest a way to read any EDOPS signature: as two complementary layers of information. A baseline layer places the basin in the global distribution—its position on the climate gradient, in the human-development hierarchy, along the coastality axis. A divergence layer registers how the basin departs from default at that position—magnitude information beyond LISA class, regional decoupling between globally correlated variables, the s/u tail where upstream and local catchments differ meaningfully.

Both layers are substantive. For a basin that is environmentally typical of its global neighborhood, the baseline is the signal; the divergence layer is sparse or empty, and that sparseness is itself meaningful. For a basin that occupies a distinctive position relative to similar basins elsewhere—a settlement on an exotic river crossing arid lowlands, a productive grassland at the dry end of the global aridity distribution—the divergence layer is what distinguishes it. The three propositions are protective work: each guards against a specific reductive shortcut that would collapse the divergence layer into the baseline. Reading a signature with both layers in mind preserves what makes the basin locally distinctive without sacrificing the structural position that anchors it globally.

6. Scale: L6 vs L8 in Practice

EDOPS signatures are computed and returned at BasinATLAS Pfafstetter levels 6 and 8. The choice between them is consequential in cases: the two scales are not interchangeable, and the variable classes established in §4—*continental-gradient*, *network-topology*, *scale-dependent*, and *local-anomaly*—respond differently to scale change. This section explains why, examines the broader family of aggregation effects (the Modifiable Areal Unit Problem) under which scale choice sits, and gives practical guidance for choosing between L6 and L8 at query time.

6.1 What L6 and L8 represent

The Pfafstetter coding system hierarchically partitions land surface into nested basins. Each Pfafstetter

level subdivides a parent basin into up to 9 sub-basins—4 around the largest tributaries, 5 along the main channel—with the realized count determined by the parent's tributary structure. Basins with insufficient tributary structure subdivide into fewer, or remain as leaf basins not further partitioned, so the realized refinement ratio between adjacent levels is well below the $9\times$ theoretical maximum. Coarser Pfafstetter levels (L1–L4) correspond to major drainage divisions, finer levels (L9–L12) to small sub-catchments. EDOPS uses L6 and L8 as its working scales.

L6: ~16,000 basins globally, mean area roughly 8,300 km². Captures sub-drainage geography at a scale comparable to a small national region or a large geographic feature (the Sierra Nevada, the western Caucasus, the Drakensberg).

L8: ~190,000 basins globally, mean area roughly 700 km². Captures local geography at a scale comparable to a small county or the catchment of a tributary stream.

The realized L6 → L8 refinement ratio is $11.6\times$ —going from L6 to L8 multiplies basin count by an order of magnitude, and mean basin area divides by the same factor. The two scales return materially different answers to the same query. For some variables they return essentially the same spatial structure, and L6 and L8 are interchangeable in practice; for others they return qualitatively different patterns, and the choice of scale changes what a signature says. Which variables fall into which category, and what mechanism produces the difference, is the subject of this section.

6.2 The Modifiable Areal Unit Problem in EDOPS

The Modifiable Areal Unit Problem (MAUP) is the dependence of geographic analysis results on the spatial units chosen for aggregation—both their *scale* (how large the units are) and their *shape* (how they are bounded, including their irregularity and how they sit relative to the underlying field (Openshaw, 1984). The same gradient, the same source data, aggregated to different units, will yield different statistical summaries, different cluster patterns, and different correlation structures. MAUP is not a flaw to be corrected; it is a structural property of analysis on tessellated space, and it operates at every aggregation step. EDOPS encounters MAUP at several such steps.

Source datasets feeding EDOPS are predominantly gridded—climate fields, elevation, hydrology, and soil property rasters. BasinATLAS aggregates these grids to basin polygons at each Pfafstetter level, computing the mean (or in some cases the mode or sum) of grid cells falling within each basin. This aggregation happens before EDOPS sees the data; EDOPS inherits BasinATLAS's L6 and L8 polygons as already-aggregated units, with the aggregation logic encoded in BasinATLAS's documentation rather than in the signature itself. Adding Band T compounds this: LMR and HYDE are themselves gridded, and how they should be aggregated into EDOPS signatures—to L6/L8 basins at ingest, to the user's query unit at request time, or queried natively at the source grid—is a pending methodological decision flagged for §10. At the other end of the chain, a polygon query (a historical polity, an ecoregion, a modern administrative unit) hands EDOPS a shape that does not match any of its basin units; the typical response is a set of basin signatures whose extent overlaps the polygon, and the polygon's shape relative to the basin tessellation becomes consequential for what that set means. MAUP is a factor at every step.

Cluster cores are scale-stable; outlier counts are not. For aridity, the LISA class percentages (HH =

humid core, LL = arid core, HL = humid outlier in arid neighborhood, LH = arid outlier in humid neighborhood, NS = not significant; classes as defined in §5.1) barely move from L6 to L8 in the core classes, but the HL outlier percentage collapses by an order of magnitude:

LISA class	description	L6 (% of basins)	L8 (% of basins)	Δ
HH	humid core	4.6	3.8	-0.8
LL	arid core	30.0	30.9	+0.9
HL	humid outlier in arid neighborhood	1.4	0.2	-1.2
LH	arid outlier in humid neighborhood	0.0	0.0	0.0
NS	not significant	[to verify]	[to verify]	—

The continent-scale humid and arid cores fill the basin set proportionally at both resolutions. The HL drop reflects features like coastal strips, river corridors, and rain-shadow basins that produced visible outliers at L6 but do not scale with basin count: the geographic features themselves are fixed, the basin count surrounding them changes. Reporting outlier prevalence therefore requires care—absolute counts and per-km² densities are comparable across scales; raw percentages are not.

Scale choice affects what an EDOPS query reveals about a region. The US Pacific Northwest aridity classification (Figure 6.1, Table 6.1) illustrates the mechanism. At L6, the region has 165 basins; the HH (humid cluster core) zone is a coastal strip covering 31.5% of regional basins, and the interior east of the Cascades is uniformly NS (68.5%). No LL (arid cluster core) clusters resolve at this resolution. At L8, the region has 1,474 basins. The HH coastal zone is similar in geographic extent (now 22.7% of a larger basin total—the proportional shift reflects the new basins added in non-coastal territory, not a contracting HH zone). The L8 tessellation also resolves Cascade rain-shadow valleys as discrete sub-basins, and those sub-basins cluster into LL groups in the Columbia Plateau, the Snake River drainage, and the Owyhee—32 basins, 2.2% of the regional total, where at L6 those same locations averaged with surrounding terrain into NS units. A query at an interior location therefore receives a different LISA classification at the two scales—NS at L6, LL at L8—while the raw aridity value is similar at both. When and why that difference matters is the subject of §5.1 (which establishes that signatures need both raw values and LISA classification) and §6.5 (which operationalizes the scale choice by use case).

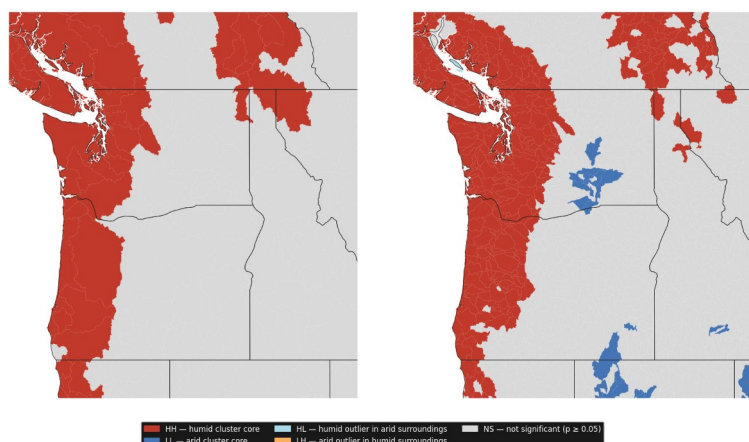


Figure 6.1 LISA clustering for aridity in US Pacific Northwest; left is L6, right L8

LISA class	description	L6 (n / %)	L8 (n / %)	Δ %
HH	humid cluster core	52 / 31.5%	334 / 22.7%	−8.9
LL	arid cluster core	0 / 0.0%	32 / 2.2%	+2.2
HL	humid outlier in arid surroundings	0 / 0.0%	1 / 0.1%	+0.1
LH	arid outlier in humid surroundings	0 / 0.0%	0 / 0.0%	0.0
NS	not significant	113 / 68.5%	1,107 / 75.1%	+6.6

Table 6.1. Aridity LISA classification, US Pacific Northwest (41.0–51.0°N, −126.0 to −113.0°W), at L6 (165 basins regionally) and L8 (1,474 basins regionally). Classifications use log-canonical aridity values, queen contiguity, 999 permutations, $p < 0.05$.

Both findings illustrate a more general principle: scale-comparison statistics should not be read as substantive without checking the mechanism. Distributional shifts across scales—for aridity, the global mean rises 11 units from L6 to L8 while the median moves by only 3—are aggregation-geometry artifacts, not real environmental shifts. Finer resolution preferentially resolves wet-extreme locations that were averaged into drier neighbors at coarser scale; the underlying climate has not changed.

In summary. The shape axis of MAUP is less characterized in CHAR’s outputs but is present in three EDOPS-specific threads. First, basin polygons are irregular—their shapes are determined by drainage, not by what most queries care about. A polity is not a basin; a city’s hinterland is not a basin; an archaeological site sits inside a basin whose shape was not designed around the site. Two basins at the same Pfafstetter level can have very different neighborhood structures (queen-contiguity counts vary widely), and the LISA-derived structural-position attributes will reflect those shape differences.

Second, polygon queries—historical polities, ecoregions, administrative units—return *sets* of basin signatures, and the relationship between the query polygon’s shape and the basin tessellation determines how clean or noisy that set is. A polygon that crops a small number of L6 basins behaves differently from one that crops hundreds of L8 sub-basins, and that is before any interior heterogeneity enters the picture; practical aggregation guidance is in §6.5.

Third, the Band T integration question raised above is itself a shape-MAUP question: choosing between basin-aggregate, query-unit-aggregate, and native-grid-query is a choice among three different aggregation shapes, each with its own MAUP profile. The methodological decision is flagged for §10.

6.3 Scale direction as variable-type diagnostic

Of the 40 variables in the univariate Moran’s I sweep, 36 show $\uparrow$ scale direction (I increases with finer resolution) and 4 show $\downarrow$ (I decreases). That 36/4 split is diagnostic..

$\uparrow$ **direction** is the *gradient-structure* default. Smaller basins sit more firmly inside individual climate, substrate (lithology and soils), or socioeconomic zones; each basin’s value tracks its immediate neighbors more closely at finer resolution. This is the expected behavior for continental-gradient and most scale-dependent variables. The largest absolute scale gains are for variables describing bounded geographic features that are smeared across coarse basins but resolve cleanly at fine basins—erosion rate ($\Delta_I = +0.181$, the largest absolute gain in the sweep), glacier coverage (+0.157), karst extent (+0.144). At the high end of the I spectrum, Band C climate variables (evapotranspiration (PET, AET), temperature, snow cover, aridity) all reach $I \geq 0.992$ at L8—effectively maximum spatial

autocorrelation. For these variables, the spatial pattern is already fully captured at L6 and L8 adds precision rather than new structural information.

↓ **direction** is the *network-topology* diagnostic. The four ↓ variables are annual discharge ($\Delta = -0.019$), minimum monthly discharge (-0.041), degree of regulation (-0.053), and silt content (-0.002 , effectively flat). The mechanism, established in §4 and §5, is intra-drainage transitions (Fig. 6.2): at L8, many more sub-basin pairs sit on opposite sides of confluence boundaries where cumulative-flow values differ by orders of magnitude, suppressing global I. Networks resolve into discontinuities at finer scale whereas gradients become smoother. Degree of regulation behaves similarly because dams are point features—at L8 a regulated basin is small and surrounded by many unregulated neighbors, so the regulated basin's value contrasts sharply with its unregulated neighbors, and Moran's I drops.

The discharge trio is a particularly informative case: three variables describing the same underlying hydrological scale show three different scale directions. Annual discharge: ↓ ($\Delta = -0.019$; standard divide effect on cumulative flow). Monthly minimum discharge: ↓ ($\Delta = -0.041$; baseflow is more sensitive to drainage geometry than annual mean flow). Monthly maximum discharge: ↑ ($\Delta = +0.024$; flood pulse propagates coherently within river corridors, so at L8 small basins within a mainstem all experience the same peak event simultaneously and form HH chains). One latent quantity, three distinct scale signatures. The trio is a reminder that "discharge" is not a single spatial pattern; the three measurements differ in scale behavior because they differ in what aspect of the flow regime they capture.

The L6/L8 choice also impacts spatial-outlier counts (HL and LH classes), which scale non-linearly. For aridity, HL and LH absolute counts grow much slower than the $11.6\times$ basin-count increase (HL grows $1.8\times$; LH stays at zero—no dry outliers within humid neighborhoods exist at either scale). For discharge, LH grows $22.5\times$ —almost double the basin-count increase. The new LH adjacencies that appear at L8 are predominantly small tributary or headwater sub-basins queen-contiguous to large-cumulative mainstem sub-basins, which is the divide effect made geographically explicit. This is why network-topology variables benefit substantively from L8 even though their global I is lower than at L6.

6.4 The contributions of L6 and L8

L6 and L8 are not better-or-worse versions of the same measurement; they are different measurements with different contributions.

L6 contributions.

- *Regional typological position.* Where a basin sits within continent-scale gradients—its position on the global aridity axis, its temperature class, its broad hydrological context. The cluster cores at L6 are reliable indicators of regional environmental character.
- *Resolution-stable cluster cores.* For large regional features (the Sahara, the Amazon rainforest, the Sahel transition zone), L6 captures the geographic extent as well as L8 does, with one-tenth the basin count.
- *Computational economy.* L6 queries are an order of magnitude smaller than L8 queries on the

same geographic extent. For exploratory or summary work, this matters.

- *Coarser unit for polygon aggregation.* When a query polygon (a polity, an ecoregion) covers a small number of L6 basins but hundreds of L8 sub-basins, L6 provides a cleaner aggregation unit unless within-polygon heterogeneity is itself the question.

L8 contributions.

- *Local character.* HL and LH spatial outliers that are geographically real but averaged into neighbors at L6. The Cascade rain-shadow basins, the loess plateau gullies, the Mediterranean orchard corridors all register at L8 in ways they cannot at L6.
- *Drainage discontinuities.* Tributary-mainstem contrasts at every confluence within a major drainage. For network-topology variables, this is the geographically informative content of the variable.
- *Bounded-feature resolution.* Glaciers, karst zones, wetlands, large urban footprints—features that have discrete geographic extents typically resolve as distinct units at L8.
- *Site-specific precision.* When a query corresponds to a specific archaeological site or settlement, L8 places the site in a basin small enough that the basin's properties approximate site-level conditions.

6.5 Scale-selection guidance by use case

The L6/L8 choice in practice depends on the query type. Four common use cases:

Single-site queries (e.g., "characterize this archaeological site"). L8 is the default. The site sits in a small enough basin that the basin's properties approximate the site's properties; L6 averages over a region that may include conditions quite different from the site's. Where the query is exploratory ("what's the general environmental class of this region?") rather than site-specific, L6 may suffice and is cheaper.

Regional summaries (e.g., "characterize the Mediterranean as an EDOPS unit"). Either scale works; the choice depends on what kind of summary is wanted. L6 gives a regional-character summary appropriate for typological work—placing a region in the continental gradient. L8 gives a within-region heterogeneity summary—surfacing the dry corridors and wet pockets that the L6 summary smooths over. For sites of interest within the region (particular cities or settlements), L8 is required to differentiate them from regional means.

Polygon aggregations (e.g., computing the EDOPS signature for a polity polygon). The polygon defines the query extent; the question is which basin-level signature aggregates into it. For polygons covering large regions (the Roman Empire at its height, the Inca domain), L6 provides natural aggregation units. For polygons covering small regions or significant within-polygon heterogeneity (city-states, river valleys, mountain kingdoms), L8 surfaces the variation that matters. A common pattern: aggregate at L8 for the analytical signature, report L6 cluster classification as regional context.

Historical queries. Scale interacts with historical validity in important ways. Continental-gradient variables—Band C climate, Band A elevation—produce essentially the same regional classification at

both scales for any query period, because the underlying patterns are stable on the relevant timescales. Network-topology variables (especially discharge) interact with both scale and history: pre-modern hydrology may differ from contemporary measurements in ways that are best read at L6 for regional flow regime and at L8 for site-specific corridor effects. Band T variables (LMR, HYDE, eVolv2k) operate at their native temporal and spatial resolutions; basin-level scale choice affects how the spatial datasets (LMR fields, HYDE land-use grids) are aggregated to the query basin, while eVolv2k provides a global-scalar forcing record that applies uniformly to any spatial unit.

A separate methodological question—how to read modern Band C and D values alongside the historical Band T reconstructions returned in the same response—is a design tension currently under discussion (see §10).

6.6 What scale does not resolve

Choice of basin scale resolves some interpretive questions but not others. Three persistent limits are worth flagging.

Within-basin heterogeneity. Even at L8, basin polygons average over internal variation. A coastal L8 basin includes both shoreline and inland areas; an alpine L8 basin includes both summit ridges and valley floors. For variables with sharp within-basin gradients (slope, soil moisture, settlement density), the L8 basin mean is still a smoothed estimate. EDOPS does not directly report sub-basin variance; users requiring within-basin detail must aggregate native grid data themselves.

Regional decoupling. Scale choice does not address the regional-behavior question raised in §4 and §5. The Mediterranean temperature $\times$ precipitation sign reversal, the Tibetan elevation $\times$ slope decoupling—these are regional departures from global patterns that hold at any basin scale. L8 reveals more local detail within the region but does not solve the regional-decoupling problem. The bivariate concordance data and the regional ESDA case studies are the relevant resources, not scale refinement.

Threshold ambiguity. Variables near class diagnostic thresholds (silt content at $\Delta_I = -0.002$, slope and elevation extremes near the continental-gradient/scale-dependent boundary) are ambiguous at any scale because their global signature is genuinely intermediate. Choosing L6 over L8 or vice versa does not resolve the ambiguity; both scales reflect a variable whose spatial structure does not commit cleanly to one typology class. These cases are flagged in the codebook and discussed in §10.

The general principle: scale is a methodological dimension that interacts with variable type, query type, and analytical purpose. There is no single "right" scale for EDOPS as a whole. The codebook annotation for each variable (typology class, scale direction, scale-sensitivity flag) carries the variable-level guidance; the use-case guidance above carries the query-level frame. §7 (Per-Band Reference) provides per-band detail on which scale serves which kind of inquiry. A fuller rubric with worked examples is planned as a follow-on to this report (§10).

7. Per-Band Reference

This section provides per-band reference detail for the EDOPS signature. The bands as a framework are introduced in §2; this section gives a compact characterization of each band's variable composition, spatial signature, historical validity, and any band-specific features a user should know when reading a

signature. Each entry follows the same structure for ease of cross-comparison.

7.1 Band A—Terrain and Geology

Variables. *Implemented:* elevation (minimum, maximum), slope, stream gradient, karst extent (%), permafrost coverage (%), lithology (categorical). *Planned:* elevation (mean), erosion rate, glacier coverage (%),

Spatial signature. I_L6 spectrum 0.66–0.92, all variables with $\uparrow$ scale direction. Predominantly *continental-gradient* (elevation mean/min/max) and *scale-dependent* (karst, glacier, erosion, slope at boundary). That class composition is gradient-and-feature: smooth elevation gradients spanning continents, plus bounded geological features (karst limestone outcrops, alpine and polar ice) that smear across L6 basins and resolve as discrete features at L8.

Distributional properties. Most Band A variables show meaningful right-skew but not extreme (skewness < 5 for most). Erosion rate is the strongest within-band scale shift in the dataset ($\Delta_I = +0.181$, the largest absolute gain across all 40 variables). Karst extent and glacier coverage show similar pattern ($\Delta_I \approx +0.15$ each). These three together demonstrate the scale-dependent class mechanism in its clearest form.

Historical validity. Stable on geological timescales, these variables describe substrate and topography that change only over millions of years. Arguably defensible for any query period.

Position attributes. `percentile` for most variables; the codebook's `position_method` column records specifics.

Categoricals. Lithology is the band's one categorical, characterized through join-count statistics rather than the Moran's I typology framework (§3).

7.2 Band B—Hydrology and Soil

Variables. *Implemented:* discharge (annual mean, monthly minimum, monthly maximum), soil texture (clay %, silt %, sand %), runoff, soil organic carbon, soil water content, groundwater depth, wetland coverage (two groups), local sub-basin river area, wetland class (categorical). *Planned:* degree of regulation, inundation maximum, lake area (%), soil organic carbon, soil water content.

Spatial signature. The band with the widest I_L6 spread in the dataset, covering 0.422 (degree of regulation) to 0.970 (soil water content). Contains all three network-topology variables (annual discharge, minimum monthly discharge, degree of regulation) and the single current local-anomaly variable (`ria_ha_ssu`). Soil and runoff variables behave as continental-gradient; the discharge cluster behaves as network-topology; the band as a whole exhibits the full range of EDOPS spatial-structure types.

Distributional properties. Discharge variables show extreme right-skew (raw skewness > 40 for annual discharge; log transform canonical). Permafrost coverage is 77% zero at L8; snow cover 57% zero; wetland and inundation also heavily zero-dominated. These distributions require careful handling: zero-inflated distributions cannot be treated as standard continuous variables for percentile calculation or comparative analysis.

Historical validity. Mixed. Soils, river network properties, and groundwater are stable on centennial scales and defensible for pre-modern queries. Discharge variables are pre-1500 valid as patterns (hydrological regimes are stable on centennial timescales) but their absolute values reflect modern measurement and require qualifying notes for historical use. Degree of regulation is strictly modern-only—dams are contemporary infrastructure.

Position attributes. `log_percentile` for the discharge cluster and river area (skewness-driven); `percentile` for soils, runoff, and groundwater.

Categoricals. Wetland class is characterized through join-count statistics.

Notable findings. The discharge trio (annual, monthly minimum, monthly maximum) exhibits three different scale directions despite measuring the same hydrological scale—a useful illustration of why "discharge" cannot be treated as a single spatial pattern (§6.3). The band contains the highest within-cluster correlations in the dataset (`discharge_yr` × `discharge_max` $r = 0.967$), but the bivariate spatial behavior demonstrates these are not interchangeable (§5.2).

7.3 Band C—Climate

Variables. *Implemented:* mean annual temperature, seasonal temperature (minimum, maximum), annual precipitation, aridity index, permafrost (%), biome and ecoregion (categoricals). *Planned:* potential evapotranspiration (PET), actual evapotranspiration (AET), climate moisture index, forest cover (%), snow cover (%).

Spatial signature. The most spatially coherent band in the dataset. PET, mean temperature, snow cover, aridity, and AET all reach $I \geq 0.992$ at L8—effectively maximum spatial autocorrelation. All Band C variables are continental-gradient with $\uparrow$ scale direction. The implication: Band C variables show a near-saturated global spatial pattern at both scales (e.g. PET $I = 0.988$ at L6, ≥ 0.992 at L8). Which regions are wet/dry, warm/cool is fully captured at either scale. L8 adds precision: a specific point's L8 basin value may differ from its L6 basin value because the L8 unit is a subset of what the L6 average covers. For broad regional typology, L6 and L8 give effectively the same answer; for site-specific queries near sharp gradients, L8 may still matter.

Distributional properties. Mean annual temperature is bimodal—a cold cluster (-5°C to 5°C ; high-latitude and high-altitude basins) and a warm cluster (20°C to 25°C ; tropical and subtropical). The trough near 10°C reflects the relative scarcity of temperate basins globally by basin count. Aridity index has a global median of 68 (semi-arid) with a right tail extending to 2,100+ in the wet tropics.

Historical validity. Sourced from contemporary measurement; can be treated as a pattern-stable proxy for some pre-modern queries. The spatial pattern (which places are wet, which dry; which warm, which cold) is generally stable on centennial-to-millennial scales, but absolute values shift. Time-resolved paleoclimate values for CE-period queries come from Band T (LMR's reconstructed anomalies relative to a modern baseline), not Band C—this is a strict separation.

Position attributes. `percentile` for continuous variables.

Categoricals. Biome and ecoregion are excluded from join-count coherence analysis by design. They are themselves spatial classifications—they encode the spatial structure being tested—so a join-count

coherence test on them would be circular and is excluded by design.

Notable findings. Band C has the tightest within-band co-variance in the dataset—the temperature cluster (temp_yr, temp_min, temp_max, temp_yr_upstream) has all six pairwise correlations exceeding $r = 0.77$, with four exceeding $r = 0.88$. However, Band C is also where regional decoupling is most prominent: the Mediterranean temperature $\times$ precipitation sign reversal, the Tibetan temperature $\times$ snow cover collapse to not-significant. For Band C, the analytical interest lies in regional variation in bivariate relationships, not in single-variable patterns (§5.2, §5.3).

7.4 Band D—Human Geography

Variables. *Implemented:* human Development Index (HDI), gross domestic product (GDP), population density, human footprint, cropland coverage (%), pasture coverage (%)⁹. *Planned:* nighttime lights index.

Spatial signature. High global I—HDI reaches 0.987/0.995 (L6/L8), GDP 0.943/0.986. All variables are continental-gradient with $\uparrow$ scale direction. . The band autocorrelates at continental scale comparably to Band C climate; contemporary human geography clusters spatially in ways that often align with environmental gradients.

Distributional properties. Right-skewed for most variables; nighttime lights is highly skewed (log transform canonical). HDI and GDP show high values concentrated in high-income regions with long left tails for low-development basins. Population density spans roughly six orders of magnitude across the global basin set.

Historical validity. Modern-only. None of the Band D variables has a defensible historical analogue at basin level—HDI is a contemporary composite metric; GDP measurement is twentieth-century; cropland and pasture extent reflect current land use, not historical. For historical queries that need anthropogenic context, Band T HYDE provides time-resolved cropland and grazing fractions; Band D should be suppressed or treated as opt-in context for historical queries. For modern or comparative queries, Band D variables also serve as correspondence targets in their own right for cultural-environmental analyses; their analytical role is not limited to historical-query suppression (§2).

Position attributes. log_percentile for nighttime lights and human footprint (skewness-driven); percentile for HDI, GDP, population density, cropland, pasture.

Notable findings. HDI $\times$ GDP shows a striking asymmetry: at L8, 8,566 basins are high-HDI / low-GDP enclaves (well-educated populations in low-cash regional economies) against 7 basins in the reverse configuration—a 1,200-fold asymmetry. Human footprint and cropland show the highest local/upstream divergence rates in the dataset ($\sim 0.6\%$), reflecting that anthropogenic land use does not respect catchment flow direction.

⁹ Cropland and pasture coverage are placed in Band D as anthropogenic land-use measures, though they sit conceptually on the boundary between human geography and biophysical character—they describe a *human transformation of environment*.

7.5 Band E—Coastality

Variables. Flow distance to marine outlet (`dist_sink_km`), outlet type (categorical: exorheic / endorheic¹⁰ / lake-terminated), downstream topological depth from coast.

Spatial signature. Small band by variable count (3 variables). Of these, only `dist_sink_km` has been processed through the univariate Moran's I sweep; it falls in the continental-gradient class with ↑ scale direction. Outlet type and topological depth have not been processed yet (`deferred-phase4` in the codebook); outlet type may ultimately be characterized as a categorical rather than a numeric variable.

Distributional properties. Flow distance to marine outlet has a heavy concentration at zero (coastal-adjacent basins) and a long right tail to interior endorheic basins thousands of kilometers from any marine sink. Outlet type is overwhelmingly exorheic (most land drains to ocean) with smaller endorheic and lake-terminated populations.

Historical validity. Stable on geomorphological timescales—coastlines and drainage outlets are physically defined and change only at the timescale of significant geomorphology (delta progradation, sea-level shift, river capture). Defensible for any query period with one major exception: active deltas (the Nile, the Yellow River, the Ganges-Brahmaputra, the Mississippi) have shifted measurably over the last several millennia, and their basin assignments may not be appropriate for queries at specific historical periods.

Position attributes. percentile for `dist_sink_km`; categorical handling for outlet type; topological depth currently `deferred-phase4`.

Notable findings. Coastality is the band most directly tied to the design commitment of treating distance-to-marine-outlet as first-class dimension (per the EDOP Prospectus). The remaining design work—full Moran's I treatment for outlet type and topological depth, and articulation of the coastality modes (riverine access, coastal-adjacency, shelf topography)—is pending and flagged for §10.

7.6 Band T—Temporal

Variables. Last Millennium Reanalysis (Last Millennium Reanalysis (LMR v2.1; reconstructed temperature and precipitation anomalies relative to a modern baseline, drought indices, on 2°×2° gridded annual fields, ~700–1900 CE reliable window).). HYDE 3.4 (cropland, grazing, irrigated and rainfed land fractions on 5'×5' gridded fields, 10,000 BCE–present). eVolv2k v4 (volcanic sulfate forcing events, hemispheric/latitudinal, ~500 BCE–1900 CE).

Spatial signature. For LMR and HYDE, native grid resolution; basin-level aggregation is noted as `deferred-phase4`. Basin-aggregated Moran's I has not been computed for Band T variables, and the four-class typology does not apply. Band T is characterized at native grid resolution: spatial coherence has been shown for HYDE epochs, the Medieval Climate Anomaly / Little Ice Age moisture dipole in LMR, and the global temperature impact of major volcanic events.

Distributional properties. LMR reliability is uneven across time and space; the proxy network is

¹⁰ Exorheic basins drain to a marine outlet (the ocean); endorheic basins are landlocked drainage systems where rivers end in inland lakes or sinks (e.g., the Aral Sea, the Caspian, Lake Eyre); lake-terminated basins drain to internal freshwater lakes that themselves connect downstream.

sparse before approximately 700 CE, particularly in the Southern Hemisphere and the tropics. HYDE BCE-period reconstructions carry substantial uncertainty (the population and land-use estimates are model-derived from assumed agricultural yields and population-density rules, not direct evidence). eVolv2k catches major events through ice-core sulfate signals; its record becomes sparse before approximately 500 BCE.

Historical validity. The only band whose values vary by query time. A B and T query for a CE-period site delivers time-resolved climate, land-use, and volcanic-forcing values calibrated for that period. The reliability constraints noted above mean that the time depth is graded: LMR reliable from approximately 700 CE forward; HYDE through the Holocene with increasing uncertainty backward in time; eVolv2k through the last ~2,500 years.

Position attributes. deferred - phase4. Position attributes for temporal variables depend on the query window, not the variable alone, and require design decisions about reference baselines that have been deferred from the CHAR phase.

Notable findings. Band T is the only band that participates in temporal scoping of signatures. Per-source reliability is treated at greater length in §8. The native-grid spatial structure of LMR and HYDE has been characterized; basin-level integration with the EDOPS signature is the substantive next phase of work and includes the position-attribute design questions flagged for §10.

8. What CHAR Does Not Establish

The CHAR phase outputs—the augmented codebook, the methodological findings of §§4–6, and the per-band reference of §7—establish a defensible foundation for EDOPS as a useful instrument. However, they do not constitute a complete validation, and they do not exhaust the questions a thoughtful user will bring to the EDOPS signature. This section names limits explicitly, separating limits of this phase of work (which subsequent phases may address) from limits inherent to the instrument or to its source data (which require different responses).

8.1 Methodological scope: exploratory, not confirmatory

CHAR’s analytical work is exploratory data analysis (EDA) and exploratory spatial data analysis (ESDA). These methods describe distributional and spatial structure; they do not test causal hypotheses, establish predictive relationships, or validate the signature against external criteria. Several consequences follow.

Moran’s I measures spatial autocorrelation—the degree to which neighboring basins share similar values—but does not address why a variable autocorrelates as it does, whether the spatial pattern is causally meaningful, or whether the variable predicts any outcome of interest. The variable typology (§4) reflects how variables autocorrelate, not what their values mean for any specific analytical purpose. A *continental-gradient* variable is one whose spatial structure is gradient-shaped; this characterization is independent of whether the variable is informative for, say, distinguishing agricultural from pastoral societies.

Bivariate Moran’s I and LISA (§5.2) measure spatial concordance between variable pairs. They do not establish that the pair captures a single underlying phenomenon, nor do they rule out that an apparently

concordant pair is driven by common confounders (latitude, continental position, climate zone). The three propositions of §5 are methodological positions about how to read the data and not tests of specific scientific claims.

Statistical significance in CHAR's ESDA work uses permutation tests with 999 replications and a fixed seed. These produce p-values floored at 0.001. For continent-scale variables with very high autocorrelation (Band C: $I \geq 0.992$), the z-scores reach hundreds—these are sample-size artifacts of the permutation method on large basin sets, not measures of effect strength. The Moran's I values themselves are the relevant comparison; z-scores and p-values are diagnostic but not load-bearing.

8.2 Validation against external criteria

CHAR has not validated signatures against ground truth, against documented historical environmental conditions, against expert classification of environmental types, or against the cultural-historical categories that EDOPS signatures will eventually be tested for correspondence with. Several specific validation pathways remain open:

Correspondence experiments. The planned comparative analyses against D-PLACE, Cliopatria, the World Heritage Cities corpus, and Chinese hydrological event records have been characterized for their sampling power (§3) but not run as yet. D-PLACE over-sampling tropical wet mountains by 3.65× and the World Heritage Cities corpus over-sampling regulated river corridors by 5.55×—both relative to the global basin population—bound what those experiments can claim about correspondence between environment and culture. Until they are conducted, the question of whether EDOPS signatures discriminate meaningfully among cultural-historical categories remains open.

Documented historical landscapes. The proposed validation against documented historical landscapes (planned for a post-CHAR phase) has not been undertaken. This is the most direct test of whether signature output corresponds to expert judgment about specific places at specific periods. Until this work is done, the alignment between signature content and historians' or geographers' judgment is asserted by the instrument's design logic rather than empirically demonstrated.

Alternative signature parameterizations. The current signature design—banded structure, variable selection, position-attribute methods, four-class typology—is one of many possible parameterizations. CHAR does not establish that this parameterization is optimal in any specific sense, nor does it benchmark the current design against plausible alternatives. The choice of L6 and L8 as working scales, the selection of 40 variables for the univariate sweep, the three-source Band T composition—each is defensible on the criteria established at design time, but none is the unique correct choice.

8.3 Per-source reliability constraints

EDOPS draws on source datasets with documented limitations. Source-level reliability has been evaluated only for LMR, as noted below; for the other three sources, EDOPS adopts the constraints reported by the source authors. The major constraints follow.

BasinATLAS. Many BasinATLAS variables are model-derived aggregations of gridded source data to basin units. Values represent the modeling and aggregation decisions of the BasinATLAS authors (Linke et al. 2019) rather than direct basin-level measurements. For variables sensitive to aggregation

method (silt fraction, soil organic carbon, glacier coverage), the basin-level value carries assumptions baked in at the aggregation step. EDOPS treats BasinATLAS values as authoritative for the purposes of signature construction.

LMR (Last Millennium Reanalysis v2.1; Tardif et al. 2019). CHAR's analysis of reconstruction-amplitude trajectory through time (finding F10.1) characterized a sparse-proxy "regression to prior" effect: where the proxy network is thin, LMR's 20 Monte Carlo runs are constrained by the long-run climatology prior, suppressing detectable anomalies regardless of underlying climate variability. The funnel-shape boundary places the operational threshold at approximately 700 CE; before that, LMR has low power to detect anomalies in any direction. EDOPS surfaces this through an `lmr_fidelity_note` that fires for queries with `year_start < 700`. Proxy density is also structurally thinner in the Southern Hemisphere and the tropics, a property of the underlying paleoclimate observation network rather than of the reconstruction method. Independent of both temporal and hemispheric constraints, the 2°×2° grid resolution is coarse relative to basin-level queries: a single LMR grid cell at the equator covers approximately 50,000 km², and many L8 basins fall within a single grid cell. Spatial smoothness of LMR output is partly a property of the reanalysis method, not of historical climate at finer scales.

HYDE 3.4 (Goldewijk et al. 2017). Historical population and land-use reconstructions are model-derived from assumed agricultural yields, population-density rules, and back-projection from known historical demographics. BCE-period reconstructions carry the largest uncertainty; the rules used to allocate population and land use in the pre-historical record have less direct evidence than the rules used for the last few centuries.

eVol2k v4 (Toohey and Sigl 2017). Ice-core sulfate signals serve as proxies for volcanic forcing. Catches major events with strong sulfate signatures but loses coverage before approximately 500 BCE. Smaller events (volcanic sulfate input < 5 Tg) are not consistently captured. The default 5 Tg threshold for "significant" events is a parameter choice, not a natural boundary.

These constraints do not invalidate Band T as a signature layer, but they do mean that a Band T value for a specific basin at a specific time carries source-level uncertainty that the position-attribute machinery does not currently quantify. §10 flags the development of explicit reliability annotations as part of Band T position-attribute design.

8.4 What signatures capture and what they do not

The EDOPS signature, overall, describes the contemporary environmental state of a basin or set of basins, with Band T providing time-resolved overlays where source data permits. Several kinds of information that a user might want from a signature are not in it.

Trajectory versus state. Signatures capture state, not trajectory. A signature for a Yellow River basin at 1200 CE reports what the basin looked like environmentally (with appropriate temporal-scoping qualifications) but does not encode the centuries of anthropogenic transformation by which the basin came to look that way. For trajectory analysis, signatures' static content must be supplemented by external knowledge of historical process.

Basin mean versus within-basin variation. In the current v0.2 implementation, signatures aggregate to basin units. Within-basin heterogeneity is not directly represented. Two settlements within the same

L8 sub-basin will receive identical signatures even if they sit in distinct microhabitats (riverside vs upland; coastal vs interior; floodplain vs terrace). Some users will note this within-basin averaging as a significant limitation, particularly for site-specific work. Whether finer Pfafstetter levels would meaningfully reduce within-unit averaging—and at what compute cost—is flagged in §10; finer basins would shrink averaging units but even L10 (~2M basins, ~75 km² mean area) does not reach the scale at which many microhabitat distinctions operate. For site-specific environmental characterization, the signature remains a starting point that the user must refine with local-scale information.

Physical environment versus cultural perception. The signature describes the physical environment as characterized by Earth-system datasets. It does not represent how a basin’s inhabitants perceived, named, classified, or used that environment. The distinction matters: physical aridity is one thing; cultural categories like “the desert” or “the steppe” are something else, suggesting boundaries and meaning according to cultural traditions that the signature does not encode. The CDOP module of the broader Computing Place initiative is the planned framework for cultural variables; EDOPS stays on the side of measurable environmental content, alongside some anthropogenic measurements.

Regional decoupling. The variable typology and the three propositions of §5 characterize global structure. As §5.2 makes explicit, regional behavior may diverge—the Mediterranean temperature × precipitation sign reversal, the Tibetan temperature × snow cover collapse to not-significant. CHAR does not characterize regional behavior comprehensively; the regional case studies that exist (Mediterranean, Monsoon Asia, Tibetan plateau, African plateau) are illustrative rather than exhaustive. Users querying signatures for sites in regions where regional decoupling is documented should consult the regional ESDA findings; users querying signatures in regions without regional case studies should treat the global typology as a working prior rather than a verified fact.

8.5 Work deferred to subsequent phases

Several substantive items have been identified as CHAR-adjacent but deferred to subsequent phases of work. Most are flagged in §10 as design decisions or operational tasks for post-CHAR development.

- **Band T basin-level aggregation.** LMR, HYDE currently operate at native grid resolution. Aggregation to basin units for full participation in the typology and position-attribute machinery is the bulk of the post-CHAR work for Band T.
- **Band E coastality modes.** Modes 2 (coastal-adjacency) and 3 (shelf topography) have not been implemented; only mode 1 (riverine access via `dist_sink_km`) is in the current signature. Outlet type and downstream topological depth remain unprocessed through the Moran’s I machinery.
- **Polygon-aggregation queries.** The API behavior for polygon queries spanning multiple basins (and across L6 and L8 levels of detail) has not been specified.
- **Areal aggregations phase.** The planned work that builds polity-level and ecoregion-level signatures in context of this basin-level instrument has not begun.
- **User-facing rubric.** A rubric for query design—guiding users through scale selection, band scoping, and signature interpretation—is planned but not drafted. Worked examples (Timbuktu, Ur, Kaifeng) from earlier CHAR exploration and the scale guidance of §6.5 suggest its shape but are not the rubric itself.

- **CDOP cultural module.** The complementary cultural module is in development but not part of the EDOPS technical track. The boundary between EDOPS and CDOP is an active design concern.

These items are scoped boundaries that subsequent work is intended to address rather than failures of CHAR. A planned October, 2026 expert review may surface additional items in this category.

8.6 The validation pathway

The aggregate effect of the limits named above is to position CHAR’s outputs as a necessary foundation for, not a substitute for, validation work. Expert review of this document is expected to address several of the design-level limits (threshold calibration, class-ontology choices, signature presentation questions). The planned correspondence experiments and worked-examples phases address several of the empirical-validation limits. Some limits—comprehensive regional behavior characterization; benchmarking against alternative signature parameterizations; quantitative reliability annotations on Band T values—remain open in directions that any single phase of work cannot fully resolve. EDOPS is being built incrementally, and §8 is a record of what that incremental construction has not yet accomplished

9. The Augmented Codebook

The augmented variable codebook (`edops_codebook_v03_draft.tsv`, 98 rows × 21 columns) records the per-variable design decisions and empirical characterizations established by CHAR. It is the operational artifact accompanying the EDOPS signature: every variable that appears in a signature has a corresponding codebook row that documents its handling. The codebook itself is a separate document released alongside this report in the project repository, where it can be versioned independently of the prose record.

The 98 rows cover all signature variables, not only the 40 continuous-numeric Band A–D variables in the univariate Moran’s I sweep. Categoricals (lithology, biome, ecoregion, wetland class, outlet type), Band T variables (LMR, HYDE, eVolv2k components), and Band E variables beyond `dist_sink_km` are all represented, with column values reflecting their distinct characterizations—**deferred-phase4** where Moran’s I-based typology does not apply, categorical-specific values for join-count-analyzed variables, and so on.

9.1 Columns added during the CHAR phase

Columns inherited from the predecessor codebook are not enumerated here. Note that `position_method` and `position_notes` refer to a future `position_score` that is not yet computed (see §10.8). The columns developed or substantially refined during CHAR include:

- **I_L6** and **I_L8**: Global Moran’s I values at Pfafstetter levels 6 and 8. Computed via queen-contiguity weights with 999-permutation significance testing. The basis for the typology assignment (§4).
- **scale_dir**: Direction of scale shift between L6 and L8 (↑, ↓, or effectively flat). Diagnostic of variable-type behavior (§4, §6).
- **typology_cluster**: Four-class typology assignment—continental-gradient,

network-topology, mixed (substantively scale-dependent; rename flagged for §10), or local-anomaly. Variables not processed through the sweep carry deferred-phase4 or categorical-specific values.

- **scale_sensitivity**: Granular flag indicating whether the variable’s spatial interpretation depends meaningfully on the choice of L6 vs L8, beyond what `scale_dir` records
- **historical_validity**: Whether the variable is valid for pre-modern queries. Values include `stable` (geological timescales), `pre-1500 valid` (pattern-stable on centennial scales), `pattern-stable proxy` (spatial pattern stable, absolute values shift), `modern-only`, and `deferred-phase4`. Independent of typology.
- **high_r_partner**: Records the variable’s partners with global Spearman $|r| \geq 0.90$. Documents the correlation structure without authorizing removal (§5.2).
- **redundancy_notes**: Free-text notes on the variable’s bivariate behavior, including spatial-vs-attribute concordance and any regional decoupling where documented.
- **distribution_notes**: Descriptive notes on the variable’s marginal distribution—skewness, zero-inflation, bi-modality, missing-data patterns.
- **position_method**: Position attribute method assigned—percentile, `log_percentile`, `rarity_rank`, `dominance_class`, or `deferred-phase4`. Determines how the variable’s value is globally calibrated in the signature (§3, §5.3). See §10.8
- **position_notes**: Free-text rationale for the position method assignment, including variable-specific handling decisions (skewness-driven log transform, zero-inflated handling deferred, regional considerations, etc.). See §10.8

The codebook is the authoritative per-variable reference. Anywhere this report discusses a variable’s typology, scale behavior, historical validity, position attribute, or co-variance/redundancy structure, the codebook is the source of truth for the specific value; this report describes the framework, the codebook records the decisions. The current version (v03) reflects CHAR phase outputs; subsequent versions will incorporate post-CHAR development work (Band T basin-level aggregation, Band E modes 2 and 3, decisions on the 14 planned variables, and threshold review per §10) together with any revisions arising from expert review.

10. Open Design Decisions

§10 catalogs the design decisions that remain open following the CHAR phase, for resolution either in future domain expert review or in post-CHAR development work. The decisions the CHAR phase has settled—choice of BasinATLAS as primary partition, L6 and L8 as working scales, the persistence-banded structure, local/upstream duality as a first-class signature feature, the four-class variable typology, position-attribute method assignments per variable, historical-validity classifications, and the codebook structure (98 rows × 21 columns)—are documented at the points where they were made across §§2–7 and are not re-enumerated here. The seven subsections that follow group open decisions by topic. Each entry carries one or more disposition tag: `[expert review]` for decisions needing expert input from the October audience; `[post-CHAR]` for operational matters scoped for the development work to follow CHAR.. Future work may surface new items, or modify existing ones.

10.1 Typology and class boundaries

Class-boundary thresholds and ambiguous variables. The four-class variable typology rests on three operational thresholds: $I_{L6} \geq 0.85$ for continental-gradient assignment; $\Delta_I < -0.010$ for network-topology assignment; outlier prevalence $> 3.0\%$ for local-anomaly assignment. These thresholds were set empirically against the distribution of values in the 40-variable sweep. Most variables cluster well away from the boundaries, but several sit close enough that minor threshold adjustments would move them between classes. Specific candidates near boundaries include silt content ($\Delta_I = -0.002$, technically $\downarrow$ but practically flat; currently assigned scale-dependent), slope and elevation extremes (near the continental-gradient / scale-dependent boundary), and karst and glacier coverage at L6 (below the $I_{L6} \geq 0.85$ line). The outlier-prevalence threshold is currently anchored on a single variable, an issue addressed separately below. The codebook currently records a single class assignment per variable, obscuring the genuine ambiguity in these cases. The decision space spans four resolutions, potentially in combination: (a) retain current point thresholds and treat ambiguity as a documentation matter only; (b) adjust thresholds based on expert judgment; (c) replace point thresholds with class-confidence ranges or explicit boundary flags; (d) add secondary class assignments to the codebook where appropriate. [expert review + post-CHAR]

Local-anomaly class single-member status. The local-anomaly class currently contains a single variable: local sub-basin river area (`ria_ha_ssu`). The class is methodologically distinct—outlier-dominated spatial structure is genuinely different from gradient or network-topology structure—but a single-member class is structurally suspicious. Two readings are available: (a) the class is real and may attract additional members as variables are added or re-characterized; (b) the class is a threshold artifact and `ria_ha_ssu` would fit better elsewhere with a different diagnostic combination. The decision affects whether the class slot is retained in the codebook ontology, eliminated, or held provisionally pending further variables. [expert review]

10.2 Variable disposition

Screen the 14 planned variables for inclusion. The CHAR sweep characterized 40 variables—26 currently implemented in the EDOPS API and 14 planned. The planned variables were included in the sweep specifically so the characterization findings could inform their final inclusion decisions. Each planned variable should be assessed against the CHAR data—Moran’s I behavior, scale direction, HL/LH outlier patterns, high-r partnerships with implemented variables—and assigned one of three dispositions: promote to implemented, continue to defer with reason stated, or cut from the codebook. A screening table (variable code, key CHAR findings, decision, brief rationale) belongs in this report or its appendix when the screening is complete. [post-CHAR]

Zero-inflated distributions. Several variables show heavy zero concentrations: permafrost 77% zero at L8; snow cover 57% zero; wetland coverage and inundation maximum similarly zero-dominated. Standard percentile-based position attributes work poorly for such distributions—a basin with permafrost coverage of zero shares position `p0` with 77% of basins globally, which is uninformative. The current codebook applies `percentile` to these variables without specialized handling. The decision is whether to develop a zero-inflated position attribute (for example, a binary “any/none” flag combined with a conditional percentile for non-zero values), accept the current limitation and

document it explicitly, or restrict these variables to non-zero analyses. The issue cuts across Bands A (permafrost), B (wetland, inundation, lake area), and C (snow cover). [post-CHAR]

10.3 Signature structure and Level additions

s/u divergence as a first-class signature feature. The signature currently reports basin-local (s) and upstream-cumulative (u) values as separate fields for each variable with the duality. §5.1 established that s/u divergence is a tail phenomenon: median divergence is zero across nine variable pairs, with strong divergence concentrated in distributional tails. For most basins, the dual reporting is redundant; where divergence fires, the divergence magnitude is itself the analytically interesting finding. The decision is whether to add an explicit divergence field (or divergence percentile) to the signature payload alongside s and u raw values, signaling to users where the duality is informative. The change has implications for signature size, JSON schema, and downstream consumer code. [expert review with user-facing input]

Seasonal/monthly resolution in the signature. The current EDOPS signature represents temperature as annual mean plus monthly minimum and maximum, and analogously for precipitation. This captures the annual range but smooths over seasonal patterns that may matter for specific research questions (agricultural calendars, seasonal mobility, settlement seasonality, transhumance patterns). The underlying basin06 and basin08 tables carry full monthly fields; whether the signature should expose monthly resolution (12 values per variable) or summary seasonal indices (season-of-maximum, summer-winter contrast, growing-season length) is a design question for signature evolution beyond the current CHAR-grounded spec. [expert review + post-CHAR]

Finer Pfafstetter levels (L9/L10). BasinATLAS provides polygons at Pfafstetter levels 1–12; EDOPS currently uses L6 and L8 only. Whether finer scales (L9 with ~600K basins, L10 with ~2M) would add enough new analytical structure to justify their cost is an open question—storage and ESDA compute scale roughly with basin count, making L10 ~10× more expensive than L8. The CHAR L6/L8 findings already constrain the answer: continental-gradient variables saturate at L8 ($I \geq 0.992$; finer scales add nothing), and network-topology variables already resolve confluences at L8 with diminishing return at finer levels. Scale-dependent and local-anomaly variables are the candidates where L9 or L10 might surface meaningfully new structure. A focused L8×L10 EDA/ESDA pass on a representative subset of variables (the discharge group, a scale-dependent candidate, and a local-anomaly variable such as degree of regulation) would clarify the cost-benefit before any commitment to finer-scale production implementation. The within-basin homogeneity limitation noted in §8.4 motivates the question from a user-experience angle alongside the methodological one, though even L10 averages ~75 km² per basin globally—finer scales reduce within-unit averaging without resolving microhabitat-scale heterogeneity. [post-CHAR]

10.4 API behavior

Neighborhood definition for point queries. When an EDOPS query is a single point, the API must choose what spatial extent the response represents. The current default returns the basin containing the point at L6 and L8. Other neighborhood definitions are possible and would serve different research questions: a distance buffer (e.g., 5km, 25km radius); a topographic zone (the local sub-basin at a finer Pfafstetter level, or a custom watershed boundary); a grid cell from a Band T data source (LMR's 2°×2°

cell or HYDE's 5-arcmin cell containing the point); a user-supplied polygon (the polygon-query case, treated separately below). The decision space includes whether to fix on one default, support a small menu of named alternatives ("basin", "grid cell", "buffer-N-km"), or accept user-defined neighborhood specifications as a query parameter. This question is cross-cutting: it shapes the polygon-aggregation question (below), the finer Pfafstetter levels question (§10.3), and Band T spatial integration (§10.6). [expert review + post-CHAR]

Query-time band selection vs analysis-time down-weighting. §2 introduced temporal scoping of signatures. For a BCE query, one might defensibly draw on Bands A, B, E, T, while qualifying Band C and suppressing Band D. The current EDOPS API behavior on this point is ambiguous: signatures could be returned in band-selected subsets (server-side filtering) or returned in full with down-weighting applied at analysis time (client-side decision). The decision affects API design, signature size on the wire, and how qualifying notes are surfaced. A reasonable resolution may be to support both modes; i.e. default to full-signature return with `_note` fields, allow band-subset queries as a convenience for users with bandwidth or interpretation constraints. [post-CHAR]

Polygon-aggregation query semantics. When an EDOPS query is a polygon (a polity boundary, an ecoregion, a study area) rather than a point or single basin, the response is underspecified in two related ways. The first is the API-semantics question: does the response aggregate to a single signature across the basins the polygon contains, or return a set of basin signatures? The second is the signature-structure question (§10.3)—if aggregated, what shape does the aggregated signature take, given that the constituent basins may straddle different typology classes, climate zones, and scales? The L6/L8 choice complicates both: the polygon may cover a small number of L6 basins (clean aggregation) but hundreds of L8 sub-basins (within-polygon heterogeneity may itself be the analytical interest). Possible behaviors include: return a set of basin signatures with metadata describing the set; aggregate at the user-specified scale only; aggregate at L8 with L6 cluster classification reported alongside; aggregate at L6 with within-polygon L8 variance reported as metadata. The decision affects API semantics, signature structure for set-responses, and the kind of analytical work polygon queries support. [expert review + post-CHAR]

Modern/historical variable interaction in historical queries. EDOPS carries thematically overlapping variables across bands: Band C provides modern climate (drawn from contemporary climatology) while LMR (Band T) provides historical climate reconstructions for 0–2000 CE; Band D provides modern anthropogenic measures (population, human footprint) while HYDE (Band T) provides historical land use and land cover. A historical query returns both—modern values for C and D, historical reconstructions from T at the queried period. How users should read these together is not yet specified: which to privilege for a given research question, how to label them in the signature payload, whether the API should filter or flag responses based on query period. This is partly a UX/UI question (signature presentation, query parameter design) and partly a methodological one (how do modern and historical values together characterize a place for a historical analysis?). Both sets of values are sound within their own frame; what's missing is design guidance for using them together. Resolution is likely tied to the broader rubric work in §10.7 and will need to be made before the API supports historical queries at production scale. [expert review + post-CHAR]

10.5 Band E completion

Band E completion. Three Band E elements remain unprocessed through the CHAR framework, all routine extensions of established methodology:

- **Outlet type characterization.** Outlet type takes three values (exorheic, endorheic, lake-terminated) and is currently uncharacterized through the Moran's I framework. The variable should be characterized as categorical through join-count statistics, the methodologically appropriate default for variables of this nature, paralleling treatment of lithology and biome.
- **Downstream topological depth.** The planned `topo_depth_from_coast` variable measures topological depth from the coast along the drainage network. It must first be computed via `next_down` DAG traversal of the HydroATLAS topology, then processed through the univariate Moran's I sweep for typology class assignment. Both steps are routine. As a planned variable, `topo_depth_from_coast` is also subject to the §10.2 screening; if promoted to implemented, the computation and sweep would follow.
- **Coastality modes 2 and 3.** Meeting Band E's stated goal of representing coastality fully requires acquiring new source data: basin-level metrics for the coastal-adjacency and shelf-topography modes do not exist in BasinATLAS and must be derived from external datasets (Natural Earth coastlines, GEBCO bathymetry, or similar). The EDOP Prospectus identifies three modes of coastality—riverine access (the drainage-network dimension currently captured by `dist_sink_km`), coastal-adjacency (proximity of the basin to a coastline regardless of drainage), and shelf topography (the continental-shelf depth structure offshore of the coastline a basin drains to). Only the first mode has variables in the current signature. Modes 2 and 3 require, in sequence: source-data identification and acquisition; design of basin-level metrics derived from these datasets; integration with the existing `dist_sink_km` framework. This is substantial design work distinct from the routine processing of the other two items in this section.

[expert review + post-CHAR; NB the expert-review aspect applies specifically to coastality modes 2 & 3]

10.6 Band T completion

Band T spatial integration with the EDOPS query model. Band T's three sources have different spatial structures and require different integration approaches: LMR (2°×2° grid) and HYDE (5-arcmin grid) are gridded data sources, while eVolv2k is a global catalog of events without a spatial grid of its own. For LMR and HYDE, a point or basin query must resolve to some spatial-extent choice—possible answer-shapes include returning the value from the single grid cell containing the query point, aggregating to basin units at L6 or L8 for parity with Bands A–E, or aggregating to a user-specified neighborhood (the broader question in the new §10.4 entry on neighborhood definition; see below). Specific design decisions for the aggregation path include time-step aggregation, masking gridded values against basin coverage, and handling of pre-700 CE LMR sparseness (§8.3). For eVolv2k, the spatial question is different—events apply at hemispheric or global scale to whatever point query falls within their zone of relevance, so the API task is event filtering by spatial relevance rather than spatial

aggregation. Band T's full participation in the EDOPS framework—typology assignment, basin-level Moran's I, position attributes—depends on these choices being made. This constitutes the bulk of post-CHAR work for Band T's spatial integration. [expert review + post-CHAR]

Band T position-attribute design. Position attributes for temporal variables are the temporal analogue of the basin-level position attributes established for static bands (§3, §5.3): in both cases the decision is what reference distribution a value is positioned against. For static bands the reference is the global basin distribution; for temporal bands it is some time-window distribution, and the choice of window carries the substantive design weight. Possible reference frames include the full historical record of the variable, a fixed reference window (a pre-industrial period, say), the contemporary climatology, or a user-specified baseline. Each choice carries interpretive implications. The temporal percentile method itself also requires design: a basin's PET in 1200 CE can be reported as "p98 against the full LMR record" or "p82 against the local-region MCA mean"—these are different statements about the same value. Related and distinct: reliability annotations for Band T values. Each source dataset carries documented reliability constraints (§8.3)—proxy network sparsity affecting LMR before approximately 700 CE; BCE-period reconstruction uncertainty for HYDE; pre-500 BCE coverage for eVolv2k. The position-attribute machinery does not currently quantify these source-level constraints in the values it reports. The decision is whether and how to surface reliability annotations in the signature payload: as flags, as confidence intervals, as text notes, or some combination. This is a substantial design problem deferred from the CHAR phase. [expert review + post-CHAR]

10.7 Documentation and user-facing materials

User-facing rubric for query design. A rubric is planned to guide users through the decisions that shape a useful EDOPS query—scale selection, band scoping, query period, interpretation of position attributes, reading of typology assignments. Design decisions for the rubric include format (decision tree, narrative guide, reference table, interactive tool), scope (which decisions are guided), target audience (specialist researchers, generalists, students), and integration with EDOPS access points (embedded in the API documentation, standalone document, repo-only). The rubric will draw on the methodological findings of §§4–6 and the per-band reference of §7, translating them into actionable guidance. The worked examples (see next item) are likely components of the rubric but not the rubric itself. The modern/historical variable interaction question (§10.4) is expected to be addressed in part through rubric guidance. [expert review + post-CHAR]

Worked examples. Earlier CHAR exploration developed worked examples for several sites—Timbuktu, Ur, Kaifeng—each illustrating a different kind of signature divergence pattern (hydrological-dominant, multi-dimensional with social-directional reversal, topographically dominant). The examples have value as documentation but are not yet structured as published material. Design decisions include which sites to include (the three above, or more, with explicit selection criteria), presentation format (signature listing with commentary, narrative explanation, comparative tables), level of detail (full signature output, selected variables, annotated highlights), and whether the examples are part of the rubric, standalone documentation, or both. [post-CHAR]

Sandbox tool extensions. The CHAR-phase choropleth interface (see §3, fourth operational output) displays individual variable distributions across basins. Extensions remain open: bivariate overlays for

variable-pair exploration, LISA classification displays for spatial-cluster visualization, regional subsetting, and time-series support for Band T variables. Design decisions for these extensions include the scope of variables and views, the user audience (internal use, collaborator-facing, fully public), and the relationship to the rubric and worked examples (is the explorer the rubric's interactive companion, or an independent tool). [post-CHAR]

10.8 Position attribute implementation

CHAR Phase 5 (§9.1) assigned a global-position method to each signature variable—percentile, log_percentile, rarity_rank, dominance_class, or deferred—and recorded the rationale in the codebook. No position scores have been computed or delivered by the API.

Implementation requires:

- For scalar variables: a CDF lookup table computed over all L8 basins per variable, used at query time to convert a raw value to a 0–1 rank
- For log-scaled variables: log1p transform applied before CDF rank
- For categoricals: a class-frequency table over L8 basins, returning the rarity fraction for the queried basin's class
- For PNV compositional shares: within-basin maximum share (dominance_class), pre-computed or derived at query time
- Band T variables: position method deferred; depends on temporal query window and LMR/HYDE design

Once implemented, the codebook gains a semantic `position_score` column in the API response, and `position_method` becomes documentation of how that score was derived rather than a prospective specification.